# Grocery Store Gloom: Food Prices as the Primary Driver of Economic Sentiment

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DS 4002:
March 25th, 2026

# Project Details

**The Core Metrics:**

- Consumer Price Index (CPI): A measure of inflation that tracks the monthly cost of a fixed "basket" of everyday goods and services (like food and shelter).
- Consumer Sentiment Index (CSI): A nationwide survey that measures how optimistic or pessimistic consumers feel about their personal finances and the broader economy.
- Broad inflation has cooled to approximately 1.4%, but consumer sentiment remains depressed due to the compounding cost of essential goods.
- When consumer sentiment drops, it reliably precedes measurable, real-world declines in aggregate household spending and GDP.

**Our Hypothesis & Value:** By using a model to identify which specific CPI categories cause the most psychological drag on sentiment, we can build a highly nuanced, predictive tool to forecast future economic pullbacks.

# Project Details

**Hypothesis:** Shelter inflation will be the primary driver of decreased consumer sentiment.

**Research Question:** Which specific component of the Consumer Price Index (CPI) acts as the primary driving factor for increased negative consumer sentiment toward the broader economy?

**Modeling Approach:** Linear Regression Model, Lasso Regression, and ARIMAX.

**Goal statement:** We will conduct a linear regression of monthly CPI components and Consumer Sentiment Index data to determine which specific categories (e.g. Shelter, Food, Transportation) contribute most significantly to decreases in consumer sentiment, measuring our success by achieving a model R-squared value of at least 0.50.

# Data Acquisition

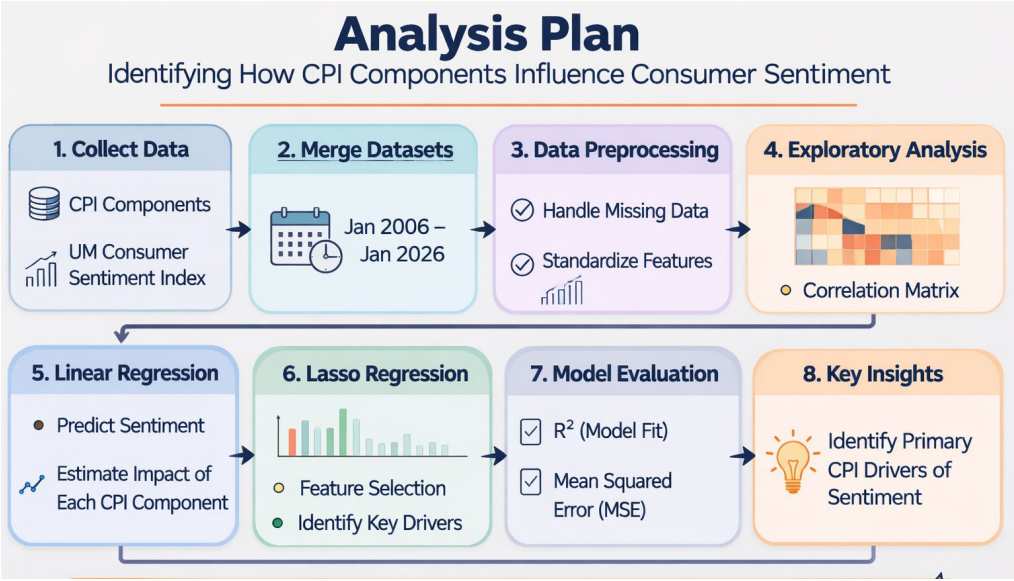
- Consumer Price Index (CPI) Data:
  - Collected monthly CPI component data from the U.S. Bureau of Labor Statistics (BLS)
- Consumer Sentiment Data
  - Obtained the University of Michigan Consumer Sentiment Index (UMCSENT) from the Federal Reserve Economic Data (FRED) database
- Dataset integration
  - Merged the CPI component dataset with the UMCSENT dataset by month and year to create a unified time series dataset for analysis
- Time Period:
  - January 2006 - January 2026 (20 years) (1 month missing - Oct 2025 due to government shutdown)
  - Eventually added more data from 1980 to 2006 from BLS website

**Data Dictionary:**

| Column | Description | Potential Responses |
|---|---|---|
| Month | The month and the year corresponding to the observation in the dataset. | Jan-06 |
| UMCSENT | The University of Michigan Consumer Sentiment Index measures consumer confidence and expectations about the economy. This is the dependent variable. | 91.2 |
| All items | 12-month percentage change in the overall Consumer Price Index (CPI), representing total inflation across all categories. | 4% |
| Food | 12-month percentage change in CPI for all food-related goods. | 2.6% |

# Analysis Plan and Justification

1. **Data Preprocessing**
   a. Restrict dataset to Jan 2006 - Sept 2025
   b. Remove missing values
   c. Standardized CPI features
2. **Feature Selection**
   a. Use **correlation matrix (|r| ≥ 0.85)** to remove multicollinearity
   b. Retain independent CPI components
3. **Baseline Model: OLS Regression**
   a. Dependent variable: **Consumer Sentiment (UMCSENT)**
   b. Evaluate using **$R^2$ + MAE**
   c. Test statistical significance (α = 0.05)
4. **Advanced Model: Lasso Regression**
   a. Perform feature selection via regularization
   b. Tune using time series cross validation
   c. Compare results with OLS

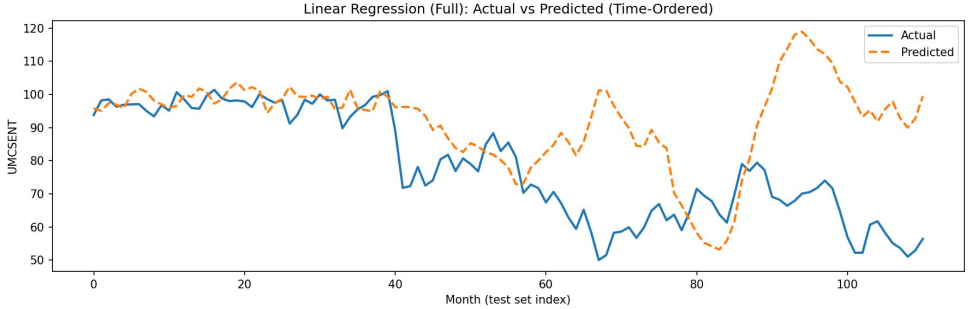


# Bias and Uncertainty Justification

The full raw dataset failed the Augmented Dickey-Fuller (ADF) test ($p = 0.1465$). Pre-COVID passed ($p = 0.0352$), suggesting the 2020 era introduced a structural bias that decoupled prices from sentiment.

Excluded categories such as "Food at Home," "Food Away from Home," and "Energy" to prevent distortion from multicollinearity

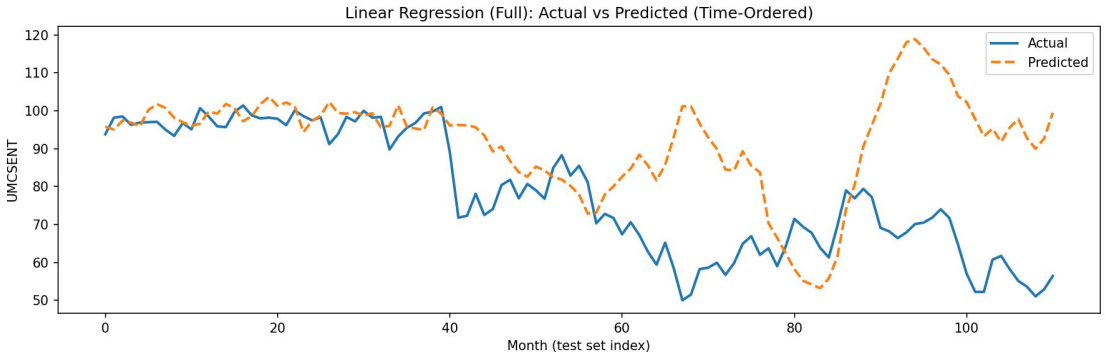
Missing Values: Removed October 2025 due to a government shutdown to maintain data integrity.




**Model Performance: $R^2$ -26.42 → $R^2$ -0.92**

# Tricky Analysis Decision

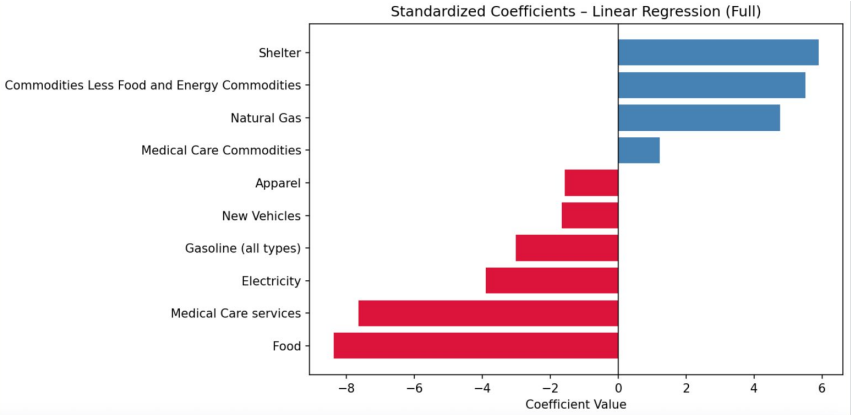
- **Problem:** Our initial 20-year model (2006-2026) yielded an $R^2$ of -26.42
- The global pandemic and subsequent years of high inflation caused a structural break in consumer sentiment with real economic data.
- We decided to rebuild the dataset and find data going back to 1980 to capture more economic cycles.
- **Validation:** Diagnostic testing proved that pre-COVID models performed significantly better than the full-horizon counterparts, confirming that the 2020 era represents a fundamental change in consumer behavior

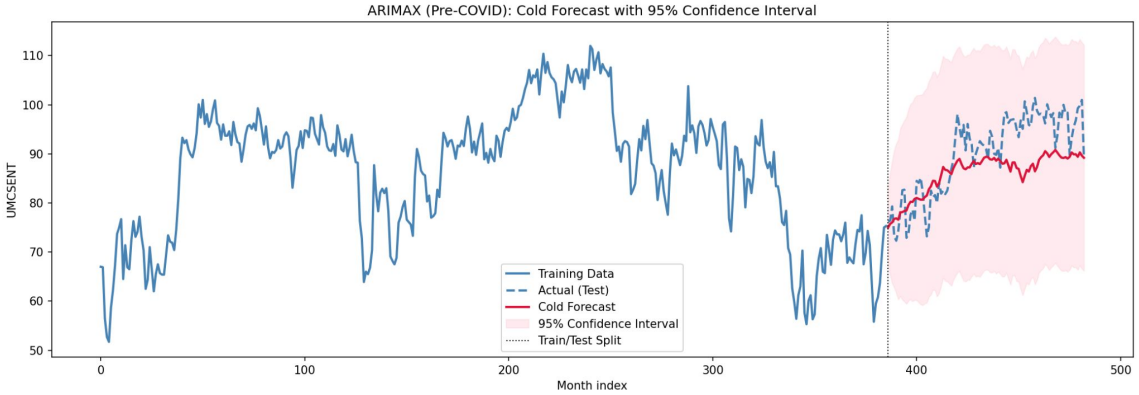



Around month 40-50 (approximately 6 years ago, or 2020), the predicted values begin to deviate significantly from what is observed.

# Results and Conclusions

- No model performed with a positive $R^2$ value when including years after the pandemic
- Food was consistently the most important negative variable across all three models, not shelter
- **Final Conclusion:** Models effectively tracked sentiment for 40 years but failed to account for the behavioral pivot that occurred after 2020.




| Model Iteration | Metric ($R^2$) (Pre - Covid) | Metric ($R^2$) (Full Model) |
|---|---|---|
| Linear Regression | 0.17 | -0.93 |
| Lasso | 0.26 | -0.17 |
| ARIMAX | 0.34 | -0.10 |

# Next Steps

### Expand Beyond CPI

- Add other macroeconomic variables:
  - Interest rates
  - Wage growth
  - Unemployment rates
  - Recession period indicators

### Segment Consumer Sentiment

- Analyze sentiment differences across different:
  - Income Groups
  - Age Groups
  - Geographic Regions

### Use Alternative Inflation Measures

- Compare measuring first order differencing for inflation
- Add lagged inflation variables

# References


[1] U.S. Bureau of Labor Statistics, *Consumer Price Index: Overview*, Sep. 9, 2024. [Online]. Available: https://www.bls.gov/cpi/overview.htm. [Accessed: Feb. 26, 2026].

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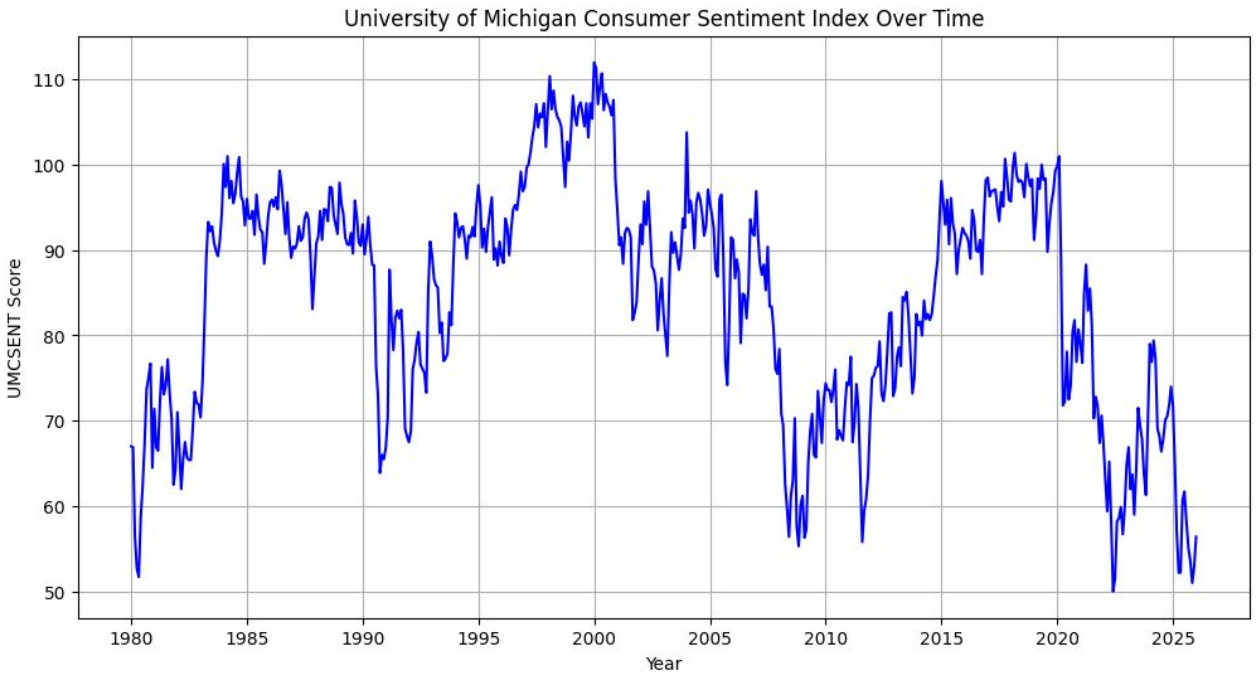
[3] University of Michigan, *Surveys of Consumers – Survey Description*. [Online]. Available: https://data.sca.isr.umich.edu/survey-description.php. [Accessed: Feb. 26, 2026].

[4] Federal Reserve Bank of St. Louis, *University of Michigan: Consumer Sentiment (UMCSENT)*. FRED. [Online]. Available: https://fred.stlouisfed.org/series/UMCSENT. [Accessed: Feb. 26, 2026].

[5] Scikit-learn developers, "1.1. Linear Models," *scikit-learn 1.8.0 documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/linear_model.html. [Accessed: Feb. 27, 2026]


Github: https://github.com/patrickzyin/DS-4002-Project-2

# Questions?


University of Michigan Consumer Sentiment Index Over Time
UMCSENT Score
110
100
90
80
70
60
50
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year